

Wettability of Solid Metals by Molten MnO-SiO₂ Slag

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An article presents results of a study of wetting of iron, nickel, and platinum substrates by molten MnO-SiO₂ slag of near-eutectic composition by the sessile-drop method in reducing atmosphere (oxygen partial pressure of 10⁻²⁰ to 10⁻¹⁸ atm). Significant MnO reduction and manganese dissolution into substrate was observed in reactions with all three substrates. Silica also was reduced in reaction with Fe and Ni, although the extent of reduction was much smaller in comparison with MnO and was not observed in reaction with Pt. The dissolution of manganese led to modification of the interface with formation of molten phase for nickel and platinum substrates. Terminal contact angles (after 240 minutes) for MnO-SiO₂ slag were as follows: for iron substrates—5 ± 2 deg at 1350 °C, 9 ± 2 deg at 1390 °C, and 6 ± 2 deg at 1450 °C; and for platinum—15 ± 2 deg at 1350 °C and 1390 °C, and 12 ± 2 deg at 1450 °C. The contact angle for the Ni- (MnO-SiO₂) system was close to zero—3 ± 2 deg at 1350 °C and 1390 °C. Modification of the metal-oxide interface as a result of reduction of oxides and adsorption of oxygen made wetting dynamic and affected interfacial properties.

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I. INTRODUCTION

IN clean steel, inclusions are formed from residual dissolved oxygen and sulfur, even at such low concentrations as 4 to 10 ppm, due to high oxygen and sulfur segregation during steel solidification. Inclusion morphology and shape (spherical, irregular crystal, or film) are affected by metal-oxide interfacial properties. In the temperature range of steel solidification (1450 °C to 1200 °C), many oxide/oxy-sulfide segregates remain liquid and spread on interdendrite surfaces, depending on the local interfacial conditions.

Wetting of solid metals by molten oxides is of interest in strip casting. Strezov *et al.*^[1] and Evans and Strezov^[2] reported that heat transfer between the molten steel and mold was affected by oxides in the FeO-MnO-SiO₂-MnS system present at the interface. When these oxides were molten, heat transfer between the steel and mold encountered very little resistance. Good wetting of the Cr-coated mold and steel interfaces by the low-melting oxide (similar to the MnO-SiO₂ eutectic) enhances heat transfer as the molten oxide will form a film across the interface. Recently, Nolli and Cramb^[3] have also shown that manganese-silicate mold coatings can improve heat-transfer rates and productivity in strip casting.

Molten oxide-solid metal systems are expected to present contrasting “wetting” behavior to the “non-wetting” character of molten metal-solid oxide systems. It can be demonstrated by considering the balance of surface energies given by Young’s equation^[4] as follows:

$$\gamma_{LV} \cos \theta = \gamma_{SV} - \gamma_{SL} \quad [1]$$

where γ_{LV} is surface tension of a liquid, γ_{SV} is surface tension of a solid, and γ_{SL} is the interfacial tension of the actual liquid-solid contact. Given the relatively high surface tensions of metals compared to oxides, the solid metal-liquid oxide configuration is expected to produce good wetting (low contact angle), while liquid metal-solid oxide systems favor a nonwetting state (high contact angle), as shown in Figure 1.

In metallurgical literature, the majority of research has been devoted to molten metal-solid oxide or molten metal-molten slag interfaces, while data on the contact angles for molten slag-solid metal system are limited. This work studied wettability and reaction of solid iron, nickel, and platinum substrates with liquid MnO-SiO₂ system with the aim to contribute to further understanding of solid metal-molten oxide interaction.

II. EXPERIMENTAL

The study is based upon *in-situ* observation of high-temperature wettability of a solid metal by low melting-temperature molten oxides (slag) by the sessile-drop method. In these experiments, the change in behavior with time and temperature of a liquid drop of slag upon a solid-metal substrate was recorded.

The experiments were conducted in a graphite furnace, schematically shown in Figure 2. The sample stage and its mounting were manufactured from graphite. A number of graphite rings were placed inside the furnace for thermal insulation. An isothermal hot zone within the furnace was measured to be larger than the sample size with a tolerance of ±1 °C at 1490 °C. The furnace was powered by a welding power supply with a peak current of 400 A. This high-current supply permitted heating rates up to 1000 °C/min.

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The water-cooled parts of the furnace were constructed from stainless steel. At one end of the chassis is the hole for insertion of a thermocouple (type O), while the other end has a glass viewing window.

A SONY* DCR-TRV18E digital video camera with a

*SONY is a trademark of Sony Corporation, Tokyo, Japan.

resolution of 720×576 pixels was used to record image sequences during the wettability experiments. A pair of Vivitar zoom lenses (supplied by Vivitar Corp., Oxnard, CA, magnification of 4 times and 1 time) were added to the camera lens to provide the correct focal length. A blue welding-goggle lens and green-tinted lens were placed between the camera and the furnace window to reduce glare from the sample at high temperatures.

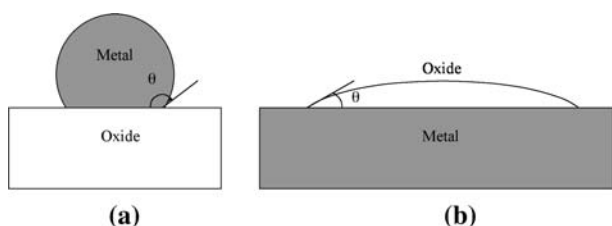


Fig. 1—Typical wetting behavior of (a) liquid metal-solid oxide and (b) solid metal-liquid oxide systems.^[5]

Iron was obtained in the form of chips (2-mm diameter). The iron was weighed out to approximately 1 g, and then melted in an alumina crucible under reducing atmosphere of Varigon** (5 vol pct hydrogen,

**Varigon is a trademark of Linde Group, Munich, Germany.

95 vol pct argon) at 1600 °C. Nickel was supplied in the form of pellets.

A sample of iron or nickel was placed inside an envelope of titanium foil, folded in at three edges. Rolling the enclosed samples pressed them to a thickness close to 1 mm, giving a diameter of approximately 15 mm. After rolling, the metal disc was polished down to a roughness of $0.05 \mu\text{m}$.

Platinum was supplied by EBS and Associates (Sydney, Australia) in the form of foil of 0.5-mm thickness. It was polished to the same finish as iron and nickel.

Chemical analysis of iron, nickel, and platinum by inductively-coupled plasma spectroscopy and LECO[†]

[†]LECO is a trademark of LECO Corporation, St. Joseph, MI.

analyzers (CS-244 carbon/sulfur and TC-436DR oxygen/nitrogen) is given in Table I.

The MnO-SiO₂ (MS) slag was prepared from the MnO and silica powder. The mixture of pure oxides was melted in a platinum-foil crucible in a muffle furnace at

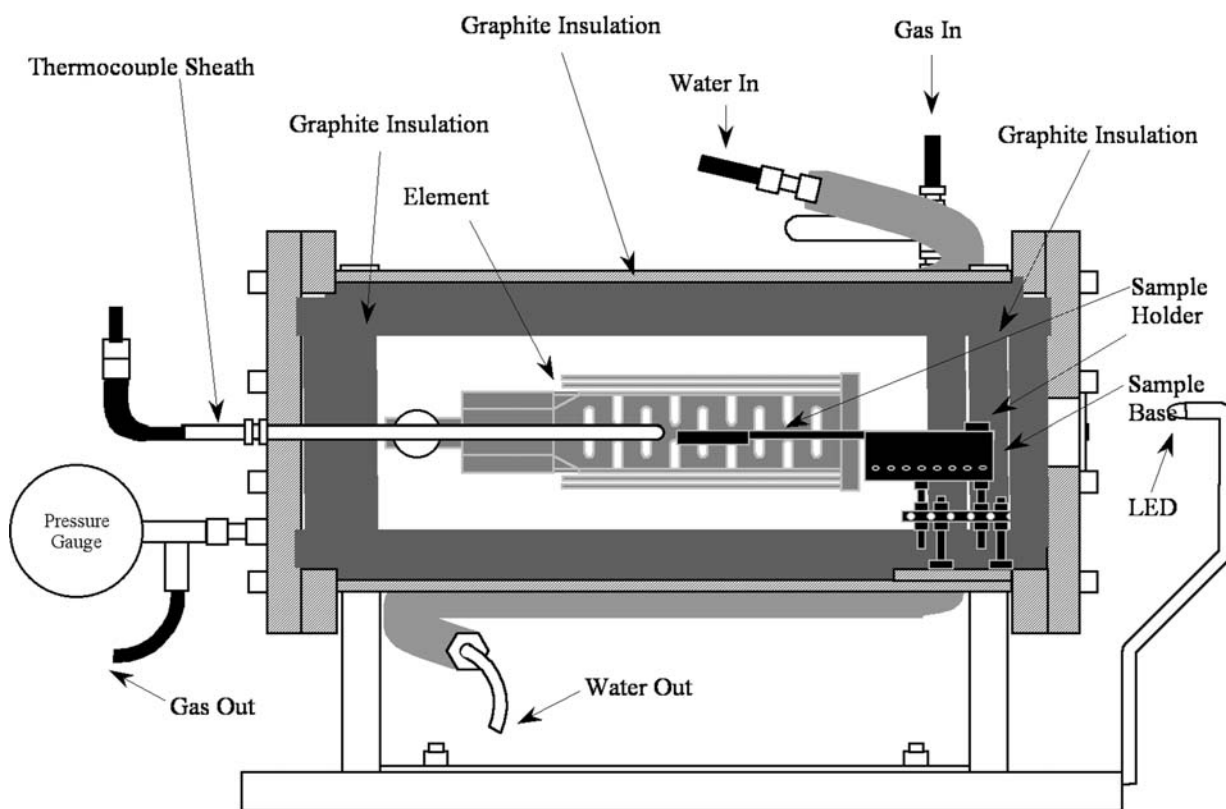


Fig. 2—Furnace schematic.

Table I. Chemical Composition of Metal Substrates (Only Elements with Concentration >10 ppm)

Element	Fe Substrate	Ni Substrate	Pt Substrate
Au	—	—	8300 ppm
C	42 ppm	100 ppm	—
Cr	<10 ppm	<0.03 wt pct	50 ppm
Cu	<10 ppm	<0.03 wt pct	100 ppm
Fe	balance	<0.03 wt pct	1200 ppm
Ni	14 ppm	balance	—
O	10 ppm	78 ppm	—
Pd	—	—	2400 ppm
Pt	—	—	balance
S	27 ppm	31 ppm	—

Table II. XRF Analysis of Slag

Compound	Concentration (Wt Pct)
MnO	58.5
SiO ₂	38.5
Fe ₂ O ₃	0.156
Na ₂ O	0.49
MgO	0.40
SO ₃	0.331
K ₂ O	0.250
CeO ₂	0.247
CaO	0.21
Al ₂ O ₃	0.14
Cr ₂ O ₃	0.095

1500 °C, and then crushed using a tungsten-carbide rod mill to particles of mass 0.01 to 0.02 g. Results of X-ray fluorescence (XRF) chemical analysis are given in Table II. The MnO-SiO₂ slag contained 58.5 wt pct MnO and 38.5 wt pct SiO₂, which is close to the eutectic composition of 61.7 wt pct MnO and 38.3 wt pct SiO₂.

Weight of substrates was approximately 1 g and of slag, approximately 0.02 g. Wetting experiments were conducted in atmospheres of Varigon (95 pct argon, 5 pct hydrogen) and argon (purity 99.999 pct). Both inlet gases were passed through copper turnings at 550 °C to remove oxygen. A stabilized zirconia oxygen probe sensor manufactured and supplied by MacDhui Engineering Pty (Melbourne, Australia) measured the oxygen partial pressure at the outlet of the furnace (uncertainty in oxygen partial pressure was $\pm 1 \times 10^{-19}$ atm).

A sample was placed on the sample stage within the furnace, which slides in and out of the hot zone. The base for the stage was aligned using a spirit level on the stage. Before experiments, the furnace was evacuated by a mechanical-vacuum pump for 15 minutes; then, the furnace was backfilled with Varigon or argon, which passed through the furnace at a flow rate of 30 mL/min.

Each experiment began with heating to 1000 °C at rapid heating rates (200 °C/min to 300 °C/min), where it was held for 30 minutes. Following this preheating, the temperature controller was then set to attain experimental temperature at 100 °C/min. Reproducibility was certified by at least two parallel runs.

The video camera captured footage at a resolution of 720 × 576 pixels. Still images were selected and stored individually using the VirtualDub program (version 1.4.10, developed by Avery Lee, sourced from www.virtualdub.org).

Measurement of surface tensions from low-angle curves observed in this work was meaningless; however, the contact angle was measured accurately. A graphical-analysis program (Grafula3 version 2.10, sourced from <http://nanodiamond.spb.ru/wesik/grafula3/english.htm>) was used to convert the images to pixel coordinates. The drop profile was marked by the user in a series of *x*- and *y*-pixel coordinates. The droplet profile was usually described by manual selection of 13 to 25 points along the edge. Each point was accurate to within ± 1 pixel (± 0.02 mm given the zoom lenses used on the camera). These selected coordinates were then subject to simple linear regression to obtain a theoretical curve matching the experimental points. The polynomial equation describing this line was used to find the value for the derivative at the limits of the curve. Taking the arc tan of these evaluations gave the contact angle for the drop. Marking the drop profile showed an error in contact-angle calculation of ± 2 deg. Experiments in which the drop profile was asymmetric were discarded. The drop radius was determined by the distance between the coordinates at the drop edges, according to calibration with a reference scale.

After experiments, samples were analyzed visually and chemically by microscopic methods. Initially, the surface appearance of the samples was recorded by the digital video camera and optical microscope (model Nikon Epiphot 200, manufactured by Nikon, Japan). Images of the cross section were recorded using the optical microscope.

In preparation for analysis by electron microscopy, the resin-mounted samples were carbon coated in a vacuum chamber. Samples were then used for scanning electron microscope (SEM) analysis with an electron-dispersive spectroscopy (EDS) attachment. The SEM machine was a Hitachi[‡] S4500 with Oxford Isis EDS.

[‡]Hitachi is a trademark of Hitachi, Ltd., Tokyo, Japan.

High-resolution diffusion profile across the interface was determined using a PHILIPS[§] CM200 with EDAX

[§]PHILIPS is a trademark of Philips Electronic Instruments Corp., Mahwah, NJ.

energy dispersive X-ray spectroscopy system.

The CAMECA^{§§} SX50 electron microprobe analyzer

^{§§}CAMECA is a trademark of CAMECA, Gennevilliers, France.

(EPMA) was used for quantitative composition analysis of the sample cross sections. The accelerating voltage was 15 kV, and the current was 20 nA. Standards used

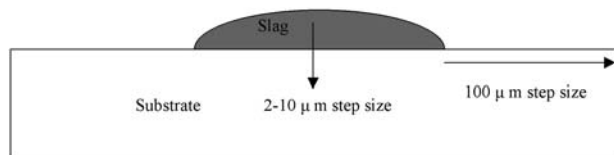


Fig. 3—Schematic of diffusion-profile paths.

were pure Fe, Ni, and Pt, and rhodonite for Mn, Si, and O. ZAF (atomic number, absorption, fluorescence) corrections were used for matrix correction. The error of measurement of manganese and silicon concentration was within 0.2 wt pct. The minimum distance between points that gives unambiguous data is approximately three microns. Profiles across the center of the slag-metal interface and parallel to the interface outward from the drop edge (as near as practical to the surface) were made with a step size of between 5 and 10 μm across and 100 μm parallel to the interface. The typical profile paths are shown schematically in Figure 3.

III. RESULTS

The following section includes experimental results on the contact angle in the molten slag-metal substrate system, obtained in isothermal experiments in Ar- H_2 and argon-gas atmospheres, slag appearance, and concentration profile across the interface.

A. Reaction of Iron Substrate with MnO-SiO_2 Slag

Experiments were conducted at temperatures of 1350 $^{\circ}\text{C}$, 1390 $^{\circ}\text{C}$, or 1450 $^{\circ}\text{C}$ for holding times of 30 or 240 minutes. The contact angle in all experiments was in the range 5 to 20 deg. The drop radius slightly decreased with time. The contact angle during isothermal experiments at 1390 $^{\circ}\text{C}$ over 240 minutes is shown in Figure 4. The contact angle measured in 30-minute experiments at different temperatures and gas atmospheres at 1390 $^{\circ}\text{C}$ is presented in Figures 5 and 6.

Figure 7 shows slag appearance after the 30-minute experiment at 1390 $^{\circ}\text{C}$. The slag color noted in Figure 7(a) is a translucent brown-orange, in contrast

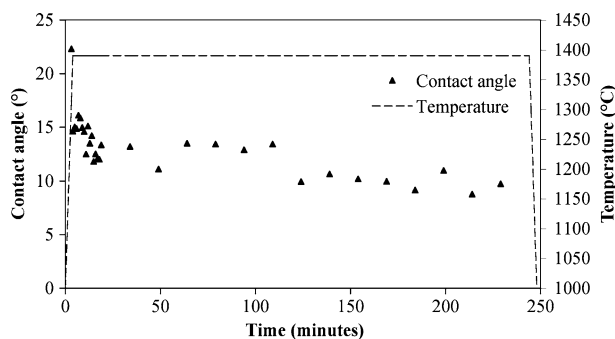


Fig. 4—Contact angle for the Fe-MS system, 240-min exposure at 1390 $^{\circ}\text{C}$.

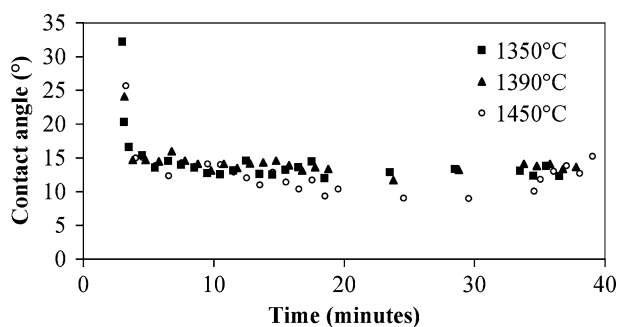


Fig. 5—Contact angle for the Fe-MS system, 30-min exposure at various temperatures.

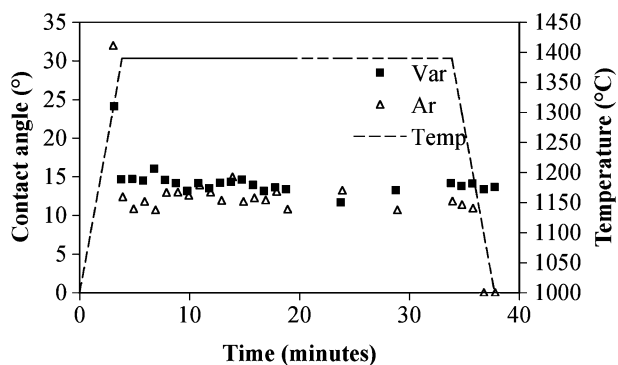


Fig. 6—Contact angle for the Fe-MS system, 30-min exposure at 1390 $^{\circ}\text{C}$ in Varigon and Ar atmospheres.

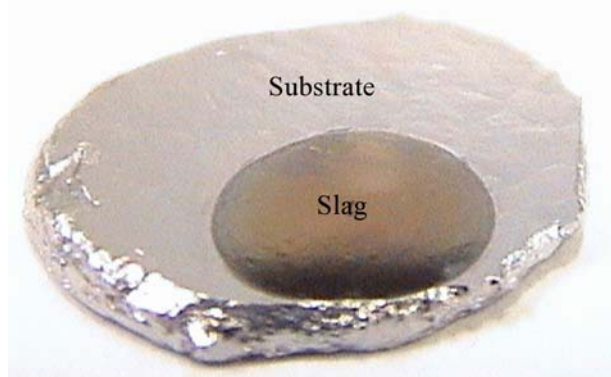
to an opaque, black original material. The microscopic view of the slag in Figure 7(b) indicates some modification of the metal surface near the contact line. The online publication provides Figure 7 in color.

The EPMA trace across the center of the interface in Figure 8 shows Mn and Si diffusion into the metal. Practically, no oxygen crossed the slag-metal interface, and diffusion of silicon diffusion into iron was only marginal. Manganese diffused into iron to a depth of 25 μm .

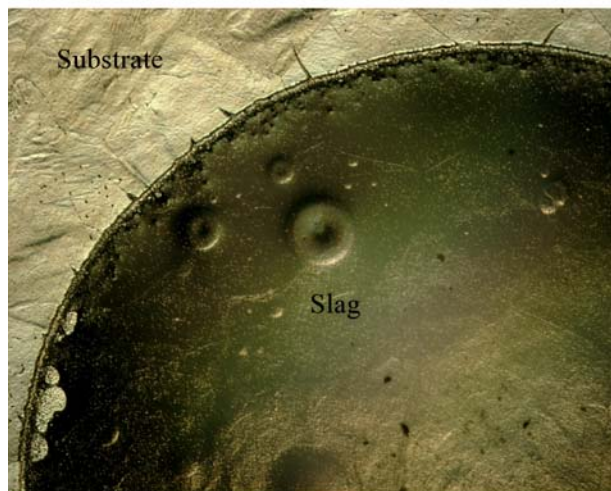
The EPMA trace parallel to the metal surface away from the interface (Figure 9) shows that Mn spread over the surface over 4 mm with a trend for decreasing concentration from approximately 1.5 wt pct. Silicon concentration was small and scattered.

B. Reaction of Nickel Substrate with MnO-SiO_2 Slag

Isothermal wetting of nickel by MnO-SiO_2 slag was studied at two temperatures (1350 $^{\circ}\text{C}$ and 1390 $^{\circ}\text{C}$) due to the lower melting point of Ni (1455 $^{\circ}\text{C}$). Nickel substrate with MS slag showed a high reactivity, characterized by very low contact angle. The contact angle measured over 30 minutes at 1390 $^{\circ}\text{C}$ is shown in Figure 10. With a longer holding time, the angle remained immeasurable because the drop profile was excessively distorted. For samples exposed for only 30 minutes, a measurable contact angle was usually restored once sample cooling began. Drop radius



(a)



(b)

Fig. 7—Appearance of the Fe-MS system after 30-min exposure at 1390 °C viewed through (a) digital camera and (b) optical microscope (50 times magnification).

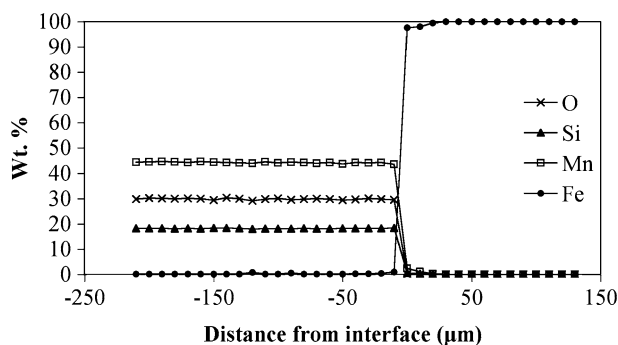


Fig. 8—Composition profile across the Fe-MS interface, 30-min reaction at 1390 °C.

increased rapidly with the lowering in angle. Effects of temperature and atmosphere were not significant.

The slag was seen to modify much of the metal substrate surface (Figure 11), which became distorted from the original flat surface. Small segments of the original slag drop were dispersed on the surface, having a pale-pink color, changed from the original black color. The online publication provides Figure 11 in color.

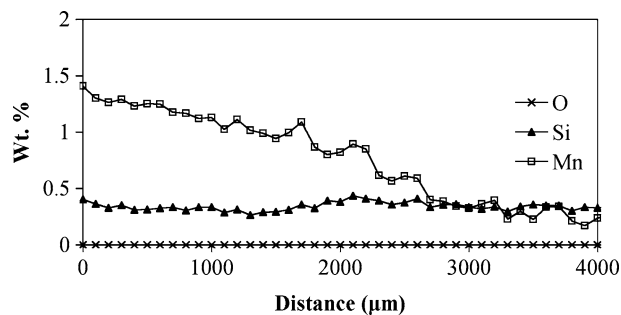


Fig. 9—Composition profile across the Fe surface, 30-min reaction with MS at 1390 °C.

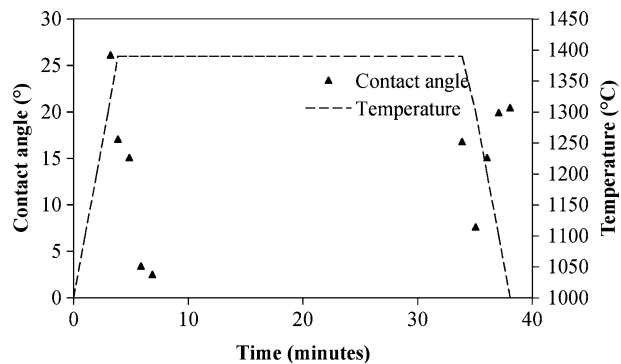


Fig. 10—Contact angle for the Ni-MS system, 30-min exposure at 1390 °C.



Fig. 11—Appearance of the Ni-MS system after 30-min exposure at 1390 °C (taken by digital camera).

The cross-sectional view (Figure 12) shows that the metal substrate has been modified to form large-scale asperities. The slag developed irregular shape and porosity. The online publication provides Figure 12 in color.

The EPMA trace across the interface (Figure 13) shows significant Mn diffusion to depths beyond 100 μm in the substrate. Si concentration in the metal was detected in lower amounts to similar depth.

Figure 14 shows that the diffusion of Mn parallel to the surface away from the slag drop occurred over a

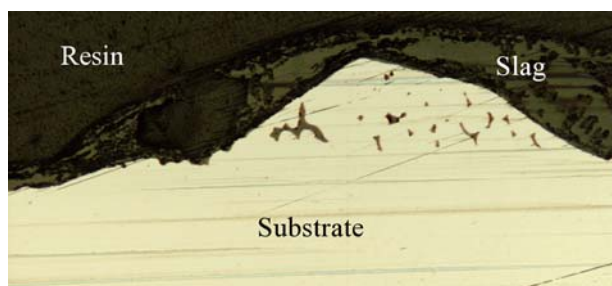


Fig. 12—Optical microscope cross-sectional view of the Ni-MS interface after 30-min exposure at 1390 °C (200 times magnification).

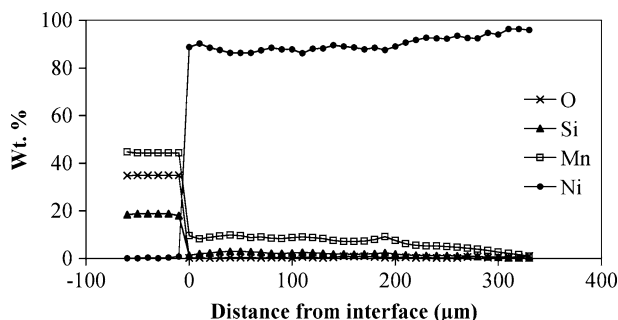


Fig. 13—Composition profile across Ni-MS interface, 30-min reaction at 1390 °C.

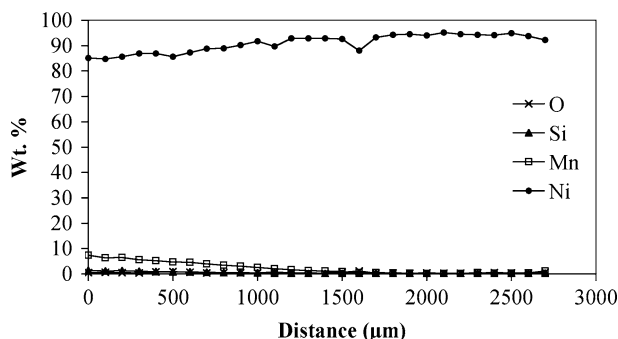


Fig. 14—Composition profile across Ni surface, 30-min reaction with MS at 1390 °C.

distance of almost 2 mm. Lesser amounts of Si were detected at the same distance.

C. Reaction of Platinum Substrate with MnO-SiO₂ Slag

Isothermal measurement of wettability of Pt substrates by MS slag was limited to 30 minutes, as longer periods led to excessive softening of the thin substrate. Contact-angle change with time is shown in Figure 15. Contact angle was approximately 15 deg over a 30-minute reaction. Drop radius remained nearly constant throughout. The effect of temperature was not significant. Evolution of gas bubbles was visually observed in the course of wetting experiments and from the slag surface appearance after experiment. The evolution

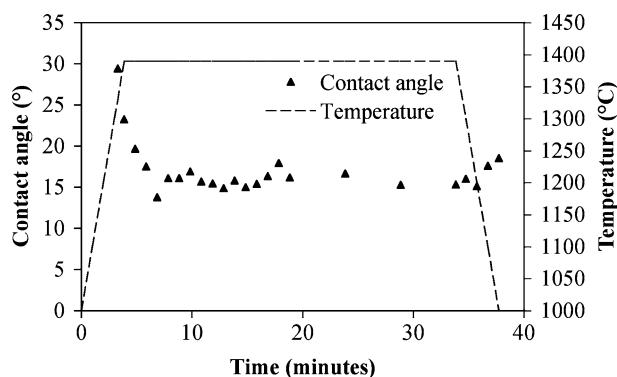


Fig. 15—Contact angle for the Pt-MS system, 30-min exposure at 1390 °C.

of gas bubbles at the platinum substrate-slag interface indicates desorption of oxygen in the process of reaction.

The slag was observed to be brown and slightly opaque, compared to the initial black appearance (Figure 16(a)). Closer examination of the slag morphology showed the surface to be made up of sac shapes in various sizes (Figure 16(b)). The fringe of dark material beyond the ring encircling the slag drop is shown in Figure 16(c). The online publication provides Figure 16 in color.

The EPMA trace across the interface (Figure 17) shows that Mn diffused in the metal to a depth of 20 μm. Dissolution of Si in the substrate was negligible.

The Mn diffusion into Pt substrate occurred over more than 1 mm in the direction away from the slag, parallel to the surface (Figure 18).

Elemental mapping at the edge of the slag/metal interface showed that Mn diffused along the substrate surface away from the slag.

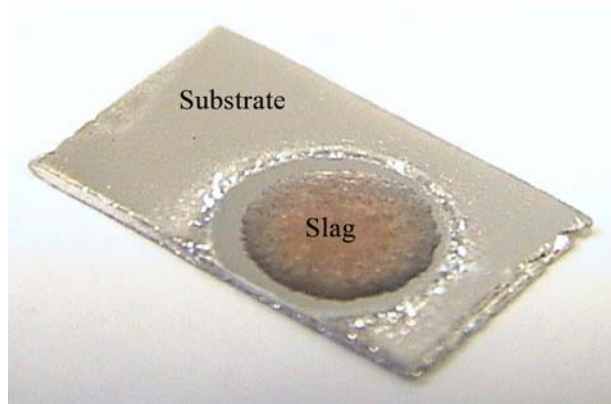
IV. DISCUSSION

A. Slag-Substrate Reaction

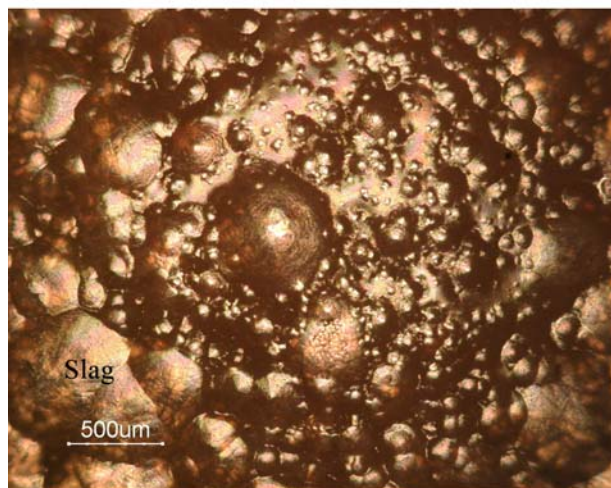
Dynamic behavior in contact angle is evidence of reactions between the metallic substrate, oxide, and gas phase. These reactions modify interfaces and change the interfacial energies.

The effect of oxygen, modifying chemistry of the interface and interfacial energies, is well recognized in wetting and other surface phenomena. A reactive or nonreactive regime of the wetting depends on the oxygen partial pressure in the furnace, oxide stability, oxygen adsorption, and solubility of oxygen and oxide-forming elements in the substrate.

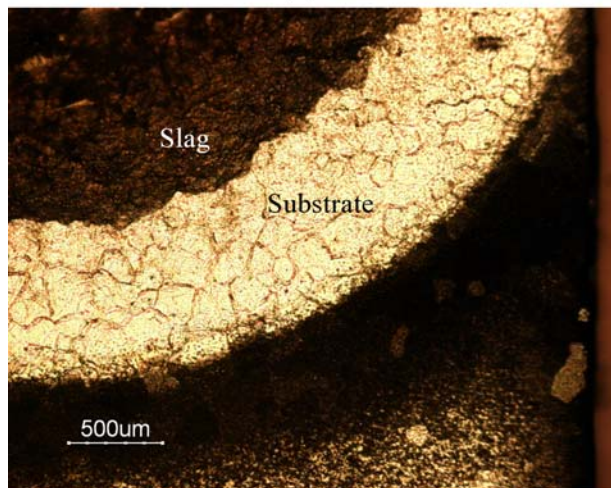
Solubility of oxygen in solid nickel, iron, and platinum is very low. Solubility of oxygen in solid nickel was found in the range of 0.01 to 0.05 at. pct,^[11,12] and rises to 1.5 at. pct upon melting.^[13] Iron has similarly low levels of solubility of oxygen in the solid state (0.0008 at. pct in γ-Fe, 0.002 at. pct in δ-Fe^[14,15]), while liquid iron at temperature close to melting point dissolves 0.06 at. pct.^[13] According to Zackay *et al.*,^[12] solubility of



(a)



(b)



(c)

Fig. 16—Surface appearance of the Pt-MS system after 30-min exposure at 1390 °C by (a) digital camera and (b) and (c) optical microscope.

oxygen in platinum is negligible, while Velho and Bartlett^[16] quote a solubility of 0.00015 at. pct.

Intake of oxygen in the wetting experiments depends on the bulk diffusion of oxygen in the metal substrates

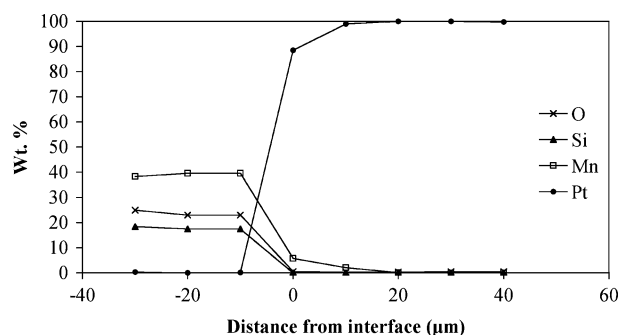


Fig. 17—Composition profile across Pt-MS interface, 30-min reaction at 1390 °C.

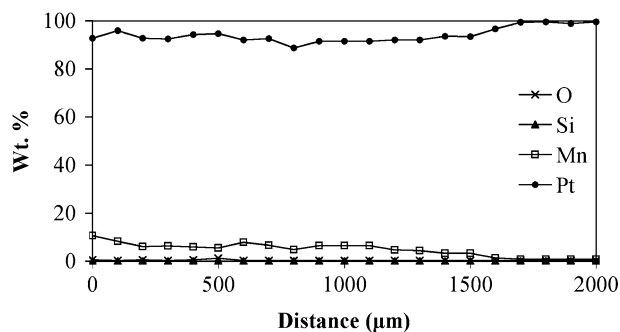


Fig. 18—Composition profile across Pt surface, 30-min reaction with MS at 1390 °C.

and the permeability (product of diffusion coefficient and solubility). At temperatures near those in the experiments, iron and nickel have relatively similar levels for oxygen-diffusion coefficient and permeability. The platinum substrate is expected to have almost no permeability for oxygen, with the diffusion coefficient being several orders of magnitude lower at the same temperature.

However, adsorption capacity of solid platinum, nickel, and iron for oxygen is known to be high. The surface activity of adsorbed oxygen on the metal substrates is highly important in determining surface tensions. Hondros^[17] estimated the surface excess of oxygen on iron at 1410 °C to be $\Gamma_O = 4 \times 10^{10}$ atoms/m². The effect of oxygen on iron surface tension was estimated at -0.1 mN/m/at. pct O. Comparable data for the other substrates were not found.

The presence of hydrogen in the furnace could influence oxygen-adsorption behavior. The solubility of hydrogen in the substrates is higher than of oxygen and followed the same trend. In the region of 1400 °C, the solubility of hydrogen in nickel is 0.04 at. pct, in iron 0.02 at. pct, and in platinum 0.01 at. pct.^[18] Platinum-group metals have a marked tendency to absorb very large volumes of hydrogen when in “black” form; platinum being the least absorbent of the group, however.^[18]

The reduction of MnO and SiO₂ can include the following reactions:

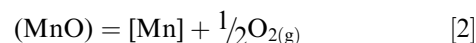
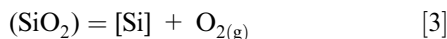
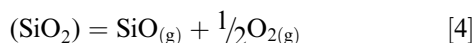


Table III. Partial Pressure of Oxygen at Metal-Oxide-Gas Equilibrium (Calculated Using Data from Reference 19)

Metal-Oxide	Equilibrium, p_{O_2} (atm)		
	1350 °C	1390 °C	1450 °C
Fe-“FeO”	6×10^{-11}	1×10^{-10}	4×10^{-10}
Mn-MnO	8×10^{-18}	3×10^{-17}	2×10^{-16}
Ni-NiO	3×10^{-6}	8×10^{-6}	3×10^{-5}
Pt-PtO ₂	1×10^5	1×10^5	2×10^5
Si-SiO ₂	8×10^{-20}	4×10^{-19}	4×10^{-18}



and



where () indicates the slag phase and [] indicates the metal phase.

The partial pressure of oxygen for the various oxides in equilibrium with pure metals is shown in Table III. Reduction of MnO and SiO₂ from the MnO-SiO₂ slag depends on the oxides stability, activity of MnO and SiO₂ in slag, and activity of Mn and Si in the metallic solution. MnO activity in the MS slag, measured in the literature^[20–23] with a mole fraction of 0.55, is approximately 0.2 (standard-state solid MnO). Silica activity in this slag calculated by the Gibbs–Duhem integration was found near 0.6.

Activity coefficients of Mn and Si in iron and nickel are well established.^[21,24,25] Sudavstova and Batalin^[26] observed strong negative deviations from Raoult’s law in Pt-Si alloys, similar to data they showed for Fe-Si and Ni-Si alloys. Abraham *et al.*,^[20] Garbers-Craig and Dippenaar,^[27] Rao and Gaskell,^[28] and Smith and Davies^[29] determined the activity coefficient of Mn in platinum dilute solution to be within the range of 10^{−2} to 10^{−4}. Schwerdtfeger and Muan^[30] found a negative deviation from Raoult’s law in the entire composition range for platinum alloyed with transition metals, finding manganese to have the strongest deviation.

With the use of the 5 pct H₂-Ar gas in the graphite furnace, partial pressure of residual oxygen was measured below 10^{−20} atm, although during the preheating at 1000 °C, it was slightly higher at 10^{−19} atm. Therefore, experimental conditions are reducing for MnO and SiO₂ in the MS slag. Oxygen released in the reduction reactions may form oxides with metals of substrates, be dissolved in the metal phase, and adsorbed on the interface or transfer to the gas phase. Oxygen solubility in solid iron, nickel, and particularly platinum is very low; it is adsorbed on the metal surface with high surface coverage but in small quantity. Therefore, oxygen can report to the gas phase.

Low-activity coefficients of manganese and silicon stimulate reduction of MnO and SiO₂ oxides with transfer of Mn and Si into the metal substrate. In accordance with activity coefficients, the degree of reduction of manganese oxide is expected to increase from Fe to Pt and further to Ni. The highest degree of

silicon reduction is also expected in reaction with nickel, followed by iron and platinum.

Visual confirmation of the reaction taking place was observed in the changing color of slag phases for most systems. The original color of MS slag was black but appeared brown-orange in most samples of iron and nickel substrates and purple-orange in platinum samples. Trace levels of reaction-product phases (metallic iron, nickel, or platinum) are known to alter the color of oxides, even on the parts per million level.^[7,31–36]

The occurrence of chemical reaction, such as reduction of MnO and SiO₂, at the interface is known to provide an additional driving force for wetting, above the balance of surface tensions.^[5,37,38] Reduction of oxides, particularly MnO, in wetting experiments causes change in slag and substrate chemistry and makes wetting dynamic.

B. Interface Modification

Reduction of manganese and silicon oxides from slags in wetting experiments was expected to modify the composition of phases. The composition profiles of slags and substrates in the vicinity of the interface, measured by EPMA, were presented in Section III. The following should be mentioned in relation to the concentration-profile measurement near the phase boundaries. The microprobe take-off volume is on the order of microns.^[9] Points taken near the interface will show false emissions of X-rays from the secondary phase.^[6] The anomaly is greatest for elements of lower atomic weight.^[10] Borom and Pask^[7] noted that this effect was significant in studying the interface of iron with sodium disilicate. In this work, the step size was 10 μm, which was assumed to be sufficient to minimize errors due to secondary X-ray fluorescence.

Change in manganese and silicon concentration in slag is related to the original elemental-slag composition, given in Table IV. Figure 19 shows the manganese-concentration profile in the reaction of metallic substrates with MnO-SiO₂ slag.

The concentration and depth of manganese dissolution in the nickel are far more profound than in iron and platinum. Platinum shows slightly higher levels of manganese near the interface than does iron, but the transfer occurs to a similar depth in both; the difference in manganese contents in iron and platinum is within the experimental error.

Change in manganese content in slag relative to its concentration in the original slag (Table IV) is the least for the iron substrate and the highest for the platinum substrate; the concentration of manganese in the slag after reaction with the platinum substrate decreased by approximately 5 wt pct, reflecting scatter

Table IV. Original Elemental Slag Composition

Element	Concentration
Mn (wt pct)	46.2
Si (wt pct)	18.8
O (wt pct)	34.9

in measurements with platinum substrates. Change in the manganese concentration in slag in reaction with the nickel substrate takes an intermediate position between platinum and iron, being closer to iron. The apparent inconsistency in data on solubility of manganese in substrates and manganese concentration in the slag can be explained by greater transfer of manganese to platinum due to increased oxygen adsorption and removal to gas phase. This was confirmed by analysis of the substrate in a direction parallel to the surface (at a depth of approximately 10 μm), as shown in Figure 20.

The effect of temperature on the manganese-concentration profile in iron was, practically, imperceptible (Figure 21), although the concentration of manganese in the slag consistently decreased with increasing temperature.

A similar effect of temperature was observed for the nickel-MnO-SiO₂ system (Figure 22); manganese-concentration profiles in nickel at 1350 °C and 1390 °C are very close, but the concentration of manganese in slag decreased with increasing temperature; the temperature effect was more pronounced than for iron substrates.

Figure 23 shows that the effect of temperature on manganese-concentration profiles in platinum in the

range of 1350 °C to 1390 °C was not significant. Manganese intake increased strongly with further increase in temperature to 1450 °C; manganese was detected in platinum to depths greater than 50 μm from the interface. Data on Mn concentration in slag in experiments with Pt were scattered and inconsistent.

Significant, if not radical, change in the nickel surface at the slag-metal interface is attributed to partial melting

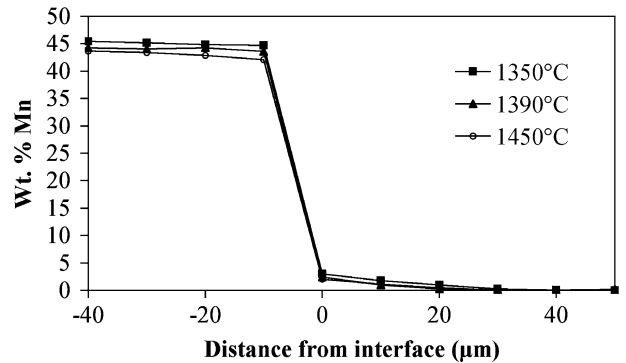


Fig. 21—Mn-concentration profile across the iron-MS slag interface after 30-min reaction at different temperatures.

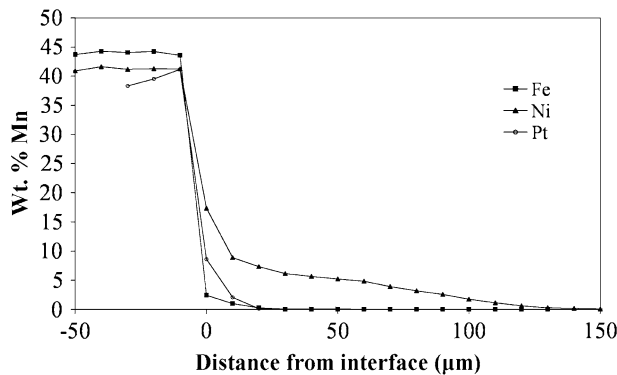


Fig. 19—Mn-concentration profile across the substrate-MS slag interface after 30-min reaction at 1390 °C.

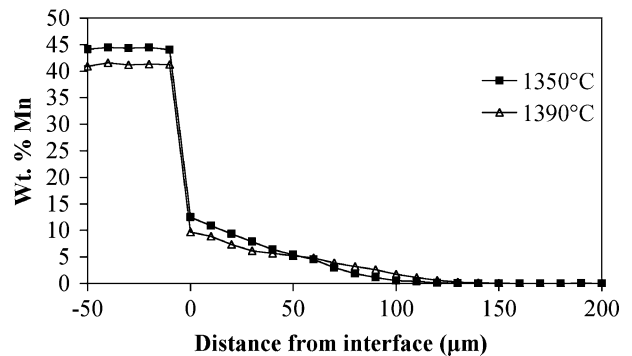


Fig. 22—Mn-concentration profile across the nickel-MS slag interface after 30-min reaction at 1350 °C and 1390 °C.

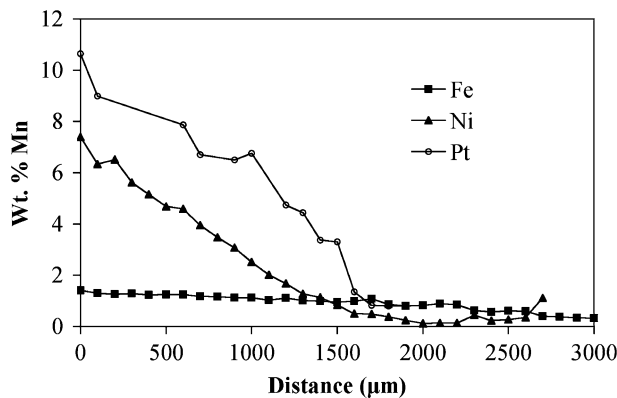


Fig. 20—Mn-concentration profile along the interface surface after 30-min reaction with MS slag at 1390 °C.

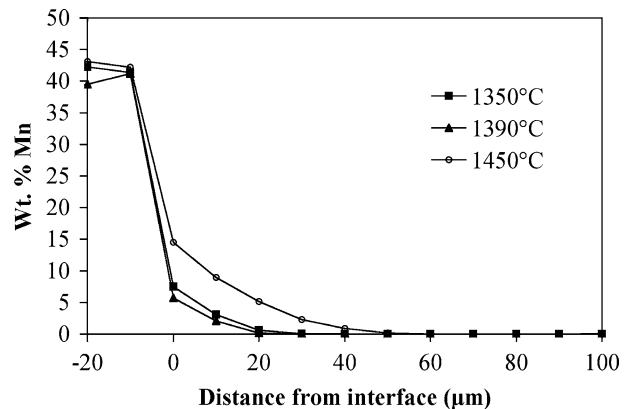


Fig. 23—Mn-concentration profile across the platinum-MS slag interface after 30-min reaction at different temperatures.

of nickel substrate as a result of dissolution of manganese in nickel, which decreases the melting temperature of the substrate. Figure 24 illustrates the movement of material down the edges of the substrate, where the left-hand edge of the substrate had a fragment of slag clinging to the side after the experiment. The online publication provides Figure 24 in color. The melting observed in some experiments at 1390 °C is supported by the Ni-Mn phase diagram.^[39]

The concentration of manganese dissolved in the nickel substrate at 1350 °C to 1390 °C achieved 10 to 12 wt pct (Figure 22). The solidus temperature of Ni-Mn alloys with 10 to 12 wt pct Mn is in the range of 1330 °C to 1380 °C, which is below the experimental temperatures; the liquidus temperature is also close to the experimental temperature. This explains the surface melting of the nickel substrate as a result of its alloying with manganese. The liquid-metal layer could be immeasurably thin. Greater extent of reaction produces a more significant liquid-metal layer.

The conversion of the slag-metal interface from liquid-solid to liquid-(partially) liquid has effects in many facets. The greater mobility in the liquid-metal enhances interfacial flux and diffusion within the metal phase. Manganese has a relatively high partial-vapor pressure. Jacob^[25] noted a loss of manganese from the Ni-Mn liquid alloy to vaporization, adding another interfacial flux. In experiments with a small slag drop, the slag dissolution into the metal caused the disintegration of the drop and deformation of the metal surface in the solidification process. In experiments with relatively large drops, the interface, covered by the slag, retained its shape.

No melting of the iron substrate is expected, according to the Fe-Mn binary-phase diagram. The map of

manganese at the surface of the iron substrate shows that the mobility of dissolved manganese was significant along the vertical-side surface of the substrate, as well as in the vicinity of the slag interface.

It follows from the binary Pt-Mn phase diagram^[39] that at 1350 °C and 1390 °C the Pt-Mn alloy with 10 to 12 wt pct Mn is solid. At 1450 °C, enrichment of the Pt-Mn alloy to approximately 15 wt pct causes it to melt. The transition to a liquid liquid-metal slag interface affects the kinetics of reaction and interfacial transport. Further dissolution of manganese into the Pt-Mn alloy to levels above 18 wt pct should lead to the alloy resolidification (β phase).

High concentration of manganese dissolved in platinum in the wetting experiments at 1450 °C agrees with literature data. Smith and Davies^[29] found 9 wt pct Mn dissolved in platinum at a partial pressure of oxygen of 10^{-7} atm at 1500 °C. Comparable figures are shown in the work of Abraham *et al.*^[40] under similar conditions. Rao and Gaskell^[41] studied the reaction of platinum with solid MnO at an oxygen partial pressure of 10^{-12} atm over a range of temperatures (1300 °C to 1600 °C). The concentration of manganese reached 12 wt pct after 80 hours of reaction. Below 1500 °C, they found formation of Pt₃Mn (approximately 9 wt pct Mn).

In experiments with manganese silicate, silica was also partially reduced from the slag. Concentration profiles of silicon in comparison with manganese for iron, nickel, and platinum substrates are presented in Figures 25 through 27.

Dissolution of silicon in nickel was much higher than in iron; it was negligible in platinum. These data are consistent with the silicon activity coefficient, which is the lowest in nickel and highest in platinum.

Formation of oxides of the substrate metal in an exchange reaction with slag cannot be excluded. Formation of Fe₂O and NiO in reduction of silicates by solid metals was observed in several studies.^[42–45] Borom and Pask^[46] termed these “adherence oxides,” given their effect on improving adhesion of slag to metal when formed at the interface in sufficient thickness. Saturation of the nickel and iron substrates with adsorbed oxygen tends to form FeO/NiO at the interface, which can dissolve into the slag. Iron and nickel oxides in the slag phase can be reduced to the metallic

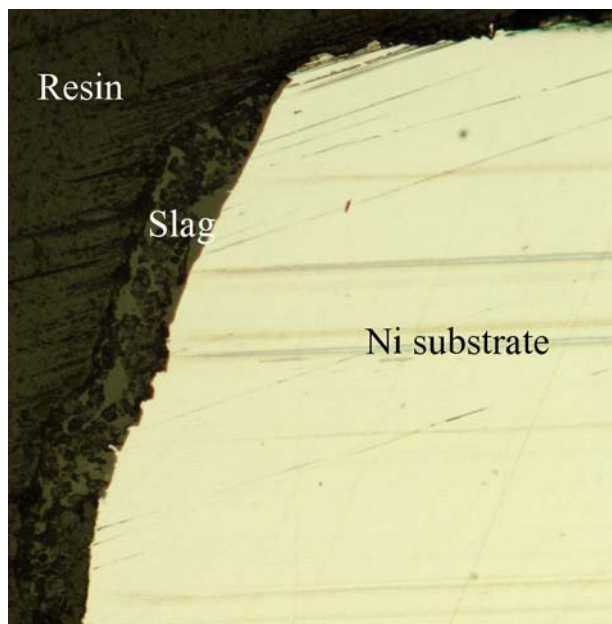


Fig. 24—Optical microscope cross-sectional view of a sample of the nickel-MnO-SiO₂ system after 30-min reaction at 1390 °C (200 times magnification).

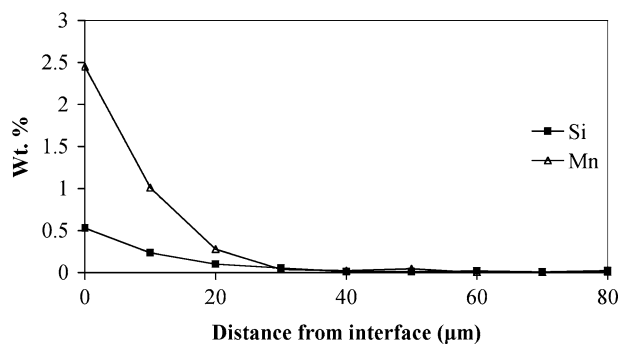


Fig. 25—Concentration profiles of Mn and Si across the iron-MnO-SiO₂ slag interface after 30-min reaction at 1390 °C.

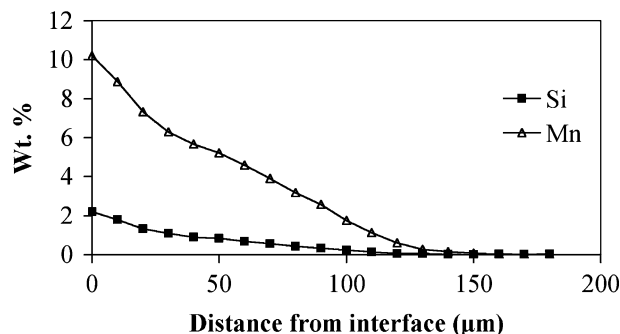


Fig. 26—Concentration profiles of Mn and Si across the nickel-MnO-SiO₂ slag interface after 30-min reaction at 1390 °C.

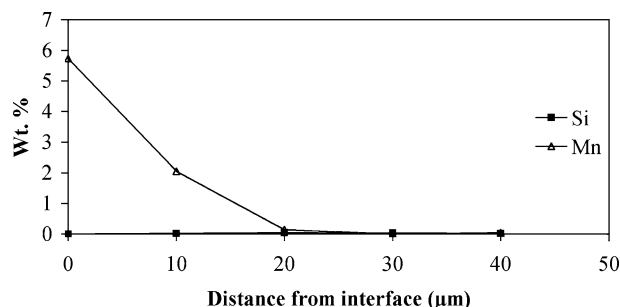


Fig. 27—Concentration profiles of Mn and Si across the platinum-MnO-SiO₂ slag interface after 30-min reaction at 1390 °C.

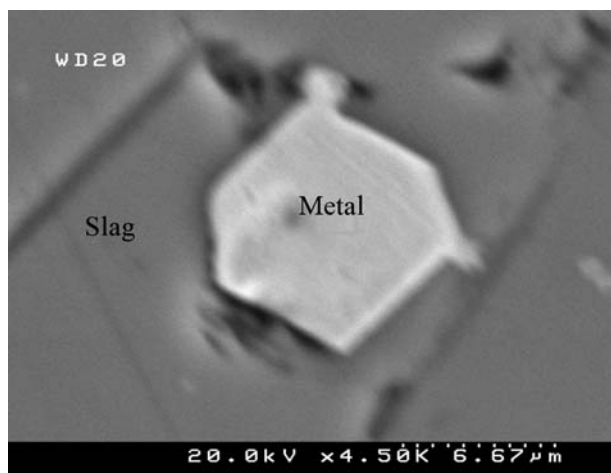


Fig. 28—SEM image of iron particle found in slag (4500 times magnification).

state under a reducing gas atmosphere. The EPMA detected only trace amounts of these metals in the slag region, most concentrated in the proximity of the interface. Further into the slag bulk, the iron and nickel were identified using SEM-EDS analysis as metallic particles (Figure 28).

The evolution of gas bubbles at platinum substrate-slag interface indicates desorption of oxygen in the process of reaction. This is sensible in light of the negligible solubility for oxygen in the platinum metal.

Distinct gas bubbles were not observed in wetting experiments with iron and nickel substrates.

V. CONCLUSIONS

In contrast to poor wetting of oxides by molten transition metals, solid metals, *i.e.*, iron, nickel, and platinum, are wetted by molten oxides very well. Measured terminal-contact angles (after 240 minutes) for the MnO-SiO₂ slag under reducing conditions were as follows: iron substrates— 5 ± 2 deg (1350 °C), 9 ± 2 deg (1390 °C), and 6 ± 2 deg (1450 °C); and platinum— 15 ± 2 deg (1350 °C and 1390 °C), and 12 ± 2 deg (1450 °C). The contact angle for the Ni-(MnO-SiO₂) system was close to zero— 3 ± 2 deg at 1350 °C and 1390 °C. The effect of temperature in the range of 1350 °C to 1450 °C for Fe and Pt and 1350 °C to 1390 °C for Ni on the contact angle was insignificant.

Experiments under argon and Varigon (Ar-5 pct H₂) atmospheres showed little effect upon wetting parameters. This reflects the strong reducing character of both atmospheres in the furnace with a graphite-heating element (partial pressure of oxygen varied from 10^{-18} to 10^{-20} atm).

Substrates reacted with MnO-SiO₂ slag with the reduction of oxides and interface modification. Reaction at the interface aided wetting in metal-slag systems, above the balance of surface tensions, leading to dynamic wetting behavior. Significant MnO reduction and manganese dissolution into the substrate was observed in reactions with all three substrates. Silica also was reduced in reaction with Fe and Ni, although the extent of reduction was much smaller in comparison with MnO and was not observed in reaction with Pt.

With reference to the phase diagrams, the dissolution of a sufficient amount of Mn led to the modification of the interface as the liquidus composition of a substrate-Mn alloy was approached. This was particularly noted for nickel substrates, with the interfacial modification often deforming the molten-drop profile so that the contact angle was undetermined.

Oxygen solubility in the substrate metals was below the detectable level, with a greater tendency to adsorb at the metal-slag interface because the reducing conditions made oxide formation on the substrates unfeasible. Adsorbed oxygen is known to be strongly surface active, lowering the surface tension of the metal substrates and favoring wetting conditions. Visual observation of oxygen desorbing as gas bubbles from the interface was made with platinum substrates.

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